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# ***Software Requirements Specification***

**For**

## **Airline Management System**

**Version 1.0**

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## Revision History

| Name                            | Date                         | Reason for Changes | Version |
|---------------------------------|------------------------------|--------------------|---------|
| Airline Management System (AMS) | 15 <sup>th</sup> March, 2022 |                    | 1.0     |

## **1. Introduction**

### **1.1. Purpose**

The main purpose of this project is to provide user friendly interface for booking flight tickets for common people and airline management for authorities. The software would provide option for checking availability of the tickets. That is important for the customers to get message if the ticket unavailable

### **1.2. Document Conventions**

There is no specific document convention for this, anyone reading by should understand easily.

### **1.3. Intended Audience and Reading Suggestions**

This document is helpful for users when they use the software and for developers and the project managers to plan their project and implement the required software. This Software Requirement Specification (SRS) document is divided into six sections.

**Section 1:** Introduction

**Section 2:** Overall Description of the Software giving information about functions, user classes, operating environment, constraints and documentation.

**Section 3:** External Interface Requirements giving a brief introduction to user, hardware, software and communications interfaces.

**Section 4:** Functional requirements of different features.

**Section 5:** Provides a list of non-functional requirements

**Section 6:** Other requirements

### **1.4. Product Scope**

This software accommodates two types of users - primary, secondary users. A primary (passenger) should be able to view the availability of flights based on his or her requirements. He or she should also be able to cancel a reservation that was made earlier. The secondary user (Airline management) should be able to view the reservation status of a particular flight and update the ticket prices of a flight. The management should be able to add a new flight to the database and also could check flight status.

### **1.5.**

#### **References**

IEEE. IEEE Std 830-1998 IEEE Recommended Practice for Software Requirements Specification. IEEE Computer Society, 1998

## **2. Overall Description**

### **2.1. Product Perspective**

Airline Management System is a self-contained product which enables a user to book flight tickets. This system will help the company to sell their flight tickets online. It also helps the passenger to check the flight timings. It also helps an airline administrator to view the reservation status of a particular or change the ticket fare by simply logging into the system.

### **2.2. Product Functions**

The following functions are provided by the software:

1. Account creation
2. Account login
3. View flight availability
4. Edit profile
5. Flight reservation request
6. Change / Cancellation of Ticket
7. Update ticket price of a particular flight
8. Add a new flight
9. Check Occupancy
10. Check profits

### **2.3. User Classes and Characteristics**

The software has the following user classes:

1. **Passenger:** The passenger can create an account by entering his or her personal information like name, age, email id and contact no. He can book a flight by entering the relevant information which includes source, destination, date and class
2. **Administrator:** The administrator can create an account by entering his ID, name, email id, contact no. He can view the status of a flight and update ticket prices.

### **2.4. Operating Environment**

**Operating System:** OS X, Ubuntu, Windows 8,10

**Language:** HTML, SQL, Django.

### **2.5. Design Implementation Constraints**

- The user must Login with their credentials to use the features

- Ticket Prices are set by airline management and do not change with rate of filling of seats
- Tickets can be booked only if a seat is available and no Waiting list of Passengers is incorporated.
- No email confirmation of ticket is sent to the passengers.
- The user is required to fill all the mandatory fields.
- Payment verification is currently unavailable.

## **2.6. User Documentation**

The Portal for the airline is very much intuitive and instructive to help the new users.

## **2.7. Assumptions and Dependencies**

- Passengers use the portal to know the fare and vacancies for a future flight trip and the airline management (referred to as admin) uses the portal to schedule and view status of flights.
- The accuracy of personal information of passengers is assumed to be true.
- Once booking is done the price of the ticket for a booking does not change with time.i.e.; no refund or extra amount is given or collected.

# **3. External Interface requirements**

## **3.1. User Interfaces**

Airline Management System uses a graphical user interface. This interface mainly uses buttons On running the software, the user will be presented with an interface containing 2 buttons User login, Administrator login.

### **Passenger login:**

- On choosing this the passenger will be presented with 3 options: login, sign up, forgot password
- By choosing login, the passenger will be asked to enter his username and password to login
- By choosing sign up the passenger will be asked to create an account and then a login page will be displayed
- Once the user has successfully logged in, he will be presented with interface containing all the functions that he can perform

### **Administrator login:**

- On choosing this the administrator will be presented with 3 options: login, sign up, forgot password
- By choosing sign up the administrator will be asked to create an account and then a login page will be displayed
- By choosing login, the administrator will be asked to enter hi username, ID, password
- Once the administrator has successfully logged in, he will be presented with interface containing all the functions that he can perform

### **3.2. Hardware Interfaces**

The minimum hardware requirements for this software are a 2.2 GHz CPU and 4 gigabytes of RAM.

### **3.3. Software Interfaces**

This software requires python 3.9 to be installed on the system and the required libraries can be downloaded if connected to the internet.

### **3.4. Communication Interfaces**

## **4. System Features**

### **4.1. Sign Up, Login, Forgot password**

#### **4.1.1 Description and Priority**

Takes the required information from the passenger to create an account and enable to login using username and password.

#### **4.1.2 Stimulus/Response Sequences**

A passenger has to login to use the features of the software. A new passenger should create an account by filling his details to sign up.

#### **4.1.3 Functional Requirements**

**Req 1:** During sign up, Ask for passenger details like name, phone number, email id, date of birth, ID (for administrator), password, and security questions

**Req 2:** Allocate space in the database

**Req 3:** Ask for username and password during login

**Req 4:** If the passenger forgets his password, he will be asked to answer the security questions provided by him during account creation.

### **4.2. Checking Availability**

#### **4.2.1 Description and Priority**

Take start and end cities along with date of departure and view list of available flights

#### **4.2.2 Stimulus/Response Sequences**

After login, if the user chooses to plan a new journey, then the system requests him to enter the following details: source, destination, and departure date.

#### **4.2.3 Functional Requirements**

**Req 1:** Ask user for aforementioned details.



**Req 2:** Access the flight schedule database and check if there is a direct flight between the two cities and display seats available as per their classes

**Req 3:** Display suitable error message if the source or destination city entered by the user is invalid. Also display an error message if there is no direct flight between the two cities.

**Req 4:** Once the user selects a flight, ask him to enter class type (business/economy/first) and number of passengers.

**Req 5:** Check if required number of seats are available or not. Display suitable error message if not available.

### **4.3. Make Reservation**

#### **4.3.1 Description and Priority**

Enable the user to book tickets in a particular flight in a particular class.

#### **4.3.2 Stimulus/Response Sequences**

If required number of seats are available in the flight, the system will proceed to make a reservation

#### **4.3.3 Functional Requirements**

**Req-1:** Ask user for number of children, number of adults and number of senior citizens

**Req-2:** Deduct the expenses of the flight tickets from the credit card linked to the user's account.

**Req-3:** Update the number of unreserved seats for the particular class on the flight.

**Req 4:** Add the flight details to the list of upcoming trips for the user.

**Req 5:** Generate and display PNR

### **4.4. Cancellation of Ticket**

#### **4.4.1 Description and Priority**

Enable the user to cancel an upcoming journey.

#### **4.4.2 Stimulus/Response Sequences**

Upon login, if the user chooses to cancel an upcoming trip, the list of upcoming trips is displayed and the user can select which flight tickets to cancel.

#### **4.4.3 Functional Requirements**

**Req-1:** Allow user to select which trip to cancel

**Req-2:** Refund the full ticket charge to the user's credit card if the cancellation is made earlier than two weeks prior to the date of journey otherwise 50 percent of the ticket charge will be refunded.

**Req-3:** Increment number of available seats on the flight.

## **4.5. Update Passenger Profile**

### **4.5.1 Description and Priority**

Enable the user to update his personal information like his phone number.

### **4.5.2 Stimulus/Response Sequences**

Choice will be given to the user to update his profile after login

### **4.5.3 Functional Requirements**

**Req-1:** Update the required field as per the user

## **4.6. Modify Ticket Price**

### **4.6.1 Description and Priority**

Allow the administrator to modify the ticket price of an upcoming flight

### **4.6.2 Stimulus/Response Sequences**

After logging in, the administrator can choose to modify the ticket price by entering the flight ID.

### **4.6.3 Functional Requirements**

**Req-1:** Ask administrator for flight ID and class type.

**Req-2:** Update the ticket price of the class type in the database.

**Req-3:** Show a confirmation message.

## **4.7. Add a new flight**

### **4.7.1 Description and Priority**

Allows the administrator to add a new flight to the database.

#### 4.7.2 Stimulus/Response Sequences

Upon login, the administrator can choose to add a new flight by entering the required information.

#### 4.7.3 Functional Requirements

**Req-1:** Ask administrator for complete details of flight (source, destination, stops, departure and arrival time, flight id, ticket fare, number of seats in each class).

**Req-2:** Update the database.

### 4.8. Check Occupancy

#### 4.8.1 Description and Priority

Allows the administrator to check occupancy of flights for a route during a period of time

#### 4.8.2 Stimulus/Response Sequences

Upon login, the administrator can choose to check profits of a route by entering the required information.

#### 4.8.3 Functional Requirements

**Req-1:** Ask administrator for complete details of route (source, destination, start time, end time)

**Req-2:** List occupancy rates for a route during a time

### 4.9. Check Profits

#### 4.8.1 Description and Priority

Allows the administrator to check net profits during a period of time

#### 4.8.2 Stimulus/Response Sequences

Upon login, the administrator can choose to check profits by entering the required information

#### 4.8.3 Functional Requirements

**Req-1:** Ask administrator for time period (start time, end time)

**Req-2:** Show profits made during a time

## **5. Other Nonfunctional requirements**

### **5.1. Performance Requirements**

- The product requires access to different databases
- The performance depends also on hardware components

### **5.2. Safety and Security Requirements**

- Only the administrator has the right to manage and make changes to the database parameters such as pricing. The system is secure and will protect the database.
- Users need to be authenticated before having access to any personal information.

### **5.3. Software Quality Attributes**

**Availability:**

The Airline Management System can be used 24 hours a day, 7 days a week

**Correctness:**

Airline Management System shall provide real time data about flight availability and pricing for a trip accurately as set by the administrator.

**Maintainability:**

Further development of code is possible and various versions can be maintained. It is easy to add additional feature to the software and modify it.

**Robustness:**

The software is made to be robust enough to have a high degree of fault tolerance. For example, the system should not crash if the user enters an invalid input. It should display a suitable error message.

**Usability:**

Airline Management System shall provide an easy-to-use graphical interface similar to existing reservation systems so that users don't need to adapt to new interface features. The interface should be interactive and easily navigable.

Any error message displayed by the software shall be polite and clear.

### **5.4. Business Rules**

*This software should be deployed only after proper contract agreement with the company which developed it. In case of any problems the user should immediately contact the project manager. The software should not be outsourced to any third party without prior permission.*

## **6. Other Requirements:**

- Licensing Requirements: Applicable
- Legal,Copyright and Other Notices: All rights reserved by our Organisation
- Applicable Standards: It should be as per Industry Standards